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QUEENSLAND'S 50-CENT FARE POLICY AND AIR POLLUTION

A Difference-in-Differences Analysis

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The Impact of Queensland's 50-Cent Flat Fare Policy on Air Pollution

A Difference-in-Differences Analysis of Brisbane vs Adelaide (2023–2025)

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Executive Summary

This report evaluates whether Queensland’s 50-cent public transport fare policy, introduced on 5 August 2024, improved air quality in Brisbane. Specifically, the analysis examines whether PM2.5 concentrations changed relative to Adelaide, which did not implement a comparable fare subsidy during the study period. From a policy perspective, the study assesses whether public transport subsidies can generate environmental benefits in addition to their transport and cost-of-living objectives.

The underlying hypothesis is that lower public transport fares encourage commuters to substitute away from private vehicle use, thereby reducing traffic-related emissions and improving air quality. To estimate this effect, a Difference-in-Differences framework is applied using daily station-level data from Brisbane and Adelaide between August 2023 and August 2025. The analysis controls for weather conditions, fuel prices, station fixed effects, and date fixed effects.

A simple before-and-after comparison suggests that pollution increased following the introduction of the subsidy. However, once weather conditions and other confounding factors are taken into account, the estimated policy effect becomes statistically insignificant. The results therefore do not provide robust evidence that the fare subsidy reduced PM2.5 concentrations during the first year following implementation.

Overall, the findings suggest that any environmental benefits of the policy were either too small to detect, offset by other factors, or had not yet materialised within the study period. Further evaluation using a longer post-policy horizon and additional environmental indicators would be required before drawing stronger conclusions regarding the policy’s environmental effectiveness.

Table of contents

Executive Summary	III
1 Introduction	4
2 Market Definition & Sector Context	5
2.1 Market definition	5
2.2 Sector Overview	5
2.3 Policy Context	5
3 Economic Framework & Empirical Literature	6
3.1 Economic Framework	6
3.2 Empirical Literature	7
4 Data & Methodology	8
4.1 Data	8
4.2 Data Processing	9
4.3 Empirical Strategy (DiD)	9
4.4 Inference Challenges	10
5 Results	11
5.1 Robustness: Leave-One-Out Analysis	13
5.2 Heterogeneity Analysis	13
6 Implications & Limitations	13
6.1 Why Might the Policy Not Have Reduced Air Pollution?	13
6.2 Limitations	14
7 References	16
Appendix	19
A.1 Descriptive Statistics	19
A.2 Air Quality Monitoring Stations	19
Brisbane	19
Adelaide	20
Choice of Unit of Analysis: City Average vs Station-Level Panel	20
A.3 Robustness Checks (Leave-One-Out)	21

1 Introduction

Urban transport policies are frequently used to address a range of economic and social objectives, including reducing congestion, improving accessibility, lowering transport costs, and mitigating the negative environmental impacts associated with private vehicle use (Serebrisky et al. 2009; Guzman and Hessel 2022). Among these policies, public transport subsidies have attracted growing attention as governments seek to encourage a shift from private cars to more sustainable transport modes. By lowering the monetary cost of public transport, such policies aim to increase ridership and reduce reliance on private vehicles, potentially generating broader benefits through reduced traffic congestion, lower emissions, and improved air quality (Gohl and Schrauth 2024).

From an economic perspective, transport-related air pollution represents a negative externality because the social costs of vehicle emissions are not fully carried by individual drivers. As a result, market outcomes may involve excessive private vehicle use relative to the socially optimal level. Public transport subsidies can therefore be viewed as a policy instrument designed to alter travel behaviour and improve welfare by reducing the environmental costs associated with road transport.

On 5 August 2024, the Queensland Government introduced a 50-cent flat fare policy across the Translink public transport network (Queensland Parliamentary Counsel 2025). The policy substantially reduced the cost of using buses, trains, ferries, and trams throughout Queensland, with the stated objectives of lowering living costs and increasing public transport usage. An important policy question is whether such a subsidy can also deliver measurable environmental benefits through reductions in air pollution.

This report evaluates whether the introduction of the fare subsidy led to lower levels of fine particulate matter (PM_{2.5}) in Brisbane. Specifically, the analysis addresses the following research question:

“Did the introduction of Queensland’s 50-cent flat fare policy reduce air pollution (PM_{2.5}) in Brisbane relative to Adelaide, which did not implement an equivalent policy during the same period?”

The underlying mechanism is straightforward. By making public transport substantially cheaper, the subsidy may encourage commuters to substitute away from private vehicle use. If this behavioural response is sufficiently large, traffic-related emissions should decrease, potentially resulting in lower concentrations of PM_{2.5}.

To estimate the causal effect of the policy, the study compares pollution levels in Brisbane and Adelaide before and after the introduction of the subsidy. A Difference-in-Differences framework is employed using daily station-level data on PM_{2.5} concentrations, weather conditions, and fuel prices. The analysis further incorporates station and date fixed effects to account for unobserved differences across monitoring locations and common shocks over time.

The remainder of the report proceeds as follows. Section 2 defines the relevant market and provides an overview of the urban transport sector and policy context. Section 3 reviews the economic framework and relevant literature. Section 4 describes the data and empirical methodology. Section 5 presents the results of the analysis, while Section 6 discusses the implications and limitations of the findings before offering concluding recommendations.

2 Market Definition & Sector Context

This section defines the relevant market, outlines key characteristics of the urban transport sector, and provides the policy context for the analysis. Understanding these elements is essential to interpret the mechanisms through which the fare subsidy may affect air pollution and to motivate the empirical strategy.

2.1 Market definition

The relevant market is the urban passenger transport market, in which public transport (buses, trains, ferries, and trams) competes with private vehicle use for commuter demand. Households choose transport modes based on cost, convenience, travel time, and accessibility. These choices have significant implications for congestion and vehicle-related air pollution.

The geographic focus of the analysis is the Brisbane metropolitan area (treated) and the Adelaide metropolitan area (control)¹. Both cities are comparable in terms of population size, urban structure, public transport networks, and climate. Adelaide did not implement a similar fare subsidy during the study period, making it a suitable control group for the difference-in-differences design. Furthermore, the substantial geographic distance between the two cities (over 2,000 km) minimises the risk of spillover effects.

The main economic actors include commuters (demand side), the Queensland Government (policy maker and funder), taxpayers, public transport operators, and public health stakeholders affected by air quality.

2.2 Sector Overview

Urban transport is closely linked to environmental outcomes. In Australia, road transport is a major source of urban air pollution, with fine particulate matter (PM_{2.5}) of particular concern due to its adverse effects on respiratory and cardiovascular health and its associated economic costs through increased healthcare expenditure, lower productivity, and reduced urban liveability (Bureau of Transport and Regional Economics 2005; Li et al. 2024; Department of Environment, Land, Water and Planning 2019).

Governments commonly intervene in this sector via infrastructure investment, pricing policies, and public transport subsidies. While pollution and congestion reduction are central goals, these policies also pursue affordability, equity, and increased public transport usage (Parry and Timilsina 2009; Centre for International Economics 2001).

Economic theory predicts that lowering the relative price of public transport should increase demand for these services and potentially reduce private vehicle use (Litman 2023). However, subsidies also carry fiscal costs, potential deadweight loss, and risks of limited effectiveness if the modal shift from cars is modest (Obeng 2010). The net environmental benefit therefore depends on demand elasticities, service quality, and prevailing conditions, making the relationship between fare subsidies and air pollution an empirical question.

2.3 Policy Context

On 5 August 2024, the Queensland Government introduced a 50-cent flat fare policy across the Translink public transport network in Southeast Queensland. Under the scheme, fares for most

¹The analysis uses data from multiple monitoring stations across suburbs in both metropolitan areas.

public transport services, including buses, trains, ferries, trams, and selected on-demand services, were reduced to a flat rate of AUD \$0.50 per journey, regardless of travel distance or fare zones. The policy was initially implemented as a six-month trial before later being made permanent in 2025 (Miles 2024; Mickelberg 2024).

The policy was introduced within a broader cost-of-living and transport policy framework. Its main objectives included improving transport affordability, increasing public transport ridership, and encouraging commuters to substitute away from private vehicle use (Queensland Parliament 2025). From an economic perspective, the policy substantially reduced the relative price of public transport compared to driving, thereby creating incentives for behavioural change in commuter transport choices.

Early evidence published by the Queensland Government and media sources suggested a significant increase in public transport demand following the introduction of the subsidy, with ridership reportedly increasing by approximately 16–18% relative to the previous year [LEKConsulting2025 Cubic Transportation Systems (2025)]. However, increased public transport usage does not necessarily imply a corresponding reduction in private vehicle usage or urban pollution levels. Consequently, the policy provides a useful quasi-natural experiment for evaluating whether a large-scale public transport subsidy can generate measurable environmental effects.

3 Economic Framework & Empirical Literature

This section outlines the economic mechanisms through which a public transport fare subsidy may influence modal choice, vehicle emissions, and urban air quality. It draws on core microeconomic concepts and reviews relevant empirical literature to frame the expected policy effects and highlight identification challenges.

3.1 Economic Framework

Urban transport markets are characterised by significant negative externalities, most notably air pollution and congestion from private vehicle use (Bertaud 2013). Private motorists typically do not internalise the full social costs of their trips (e.g., PM2.5 emissions contributing to respiratory health issues and healthcare expenditures). This creates a market failure because the private marginal cost of driving is below the social marginal cost, leading to over-consumption of car travel relative to the social optimum (Helbling 2020).

Public transport fare subsidies aim to correct this distortion by lowering the relative generalised cost of public transport. In theory, this encourages substitution away from private vehicles, potentially reducing vehicle kilometres travelled and associated emissions. The effectiveness of this mechanism depends on several factors:

- **Price elasticities:** Demand for public transport with respect to its own fare is generally inelastic in the short run (typically between -0.3 and -0.6).² This reflects habit formation, limited immediate alternatives, and the necessity of many commuting trips. Cross-price elasticities with respect to car costs are positive but often modest, requiring large fare reductions to induce meaningful modal shift (Davis 2021; Balcombe et al. 2004b).
- **Substitution patterns:** these tend to be stronger in dense urban areas with good public transport coverage and high car parking costs. However, if public transport is a weak substitute

²See, for example, Balcombe et al. (2004a) and Davis (2021).

(due to longer travel times or first/last-mile issues), ridership gains may largely come from induced demand or non-car modes rather than from private vehicles.

- **Fiscal and efficiency considerations:** subsidies impose costs on taxpayers and can generate deadweight loss. They may also encourage operational inefficiencies if operators face soft budget constraints. The net welfare effect depends on whether environmental and congestion benefits outweigh these fiscal costs.

In the context of Queensland’s 50-cent flat fare policy, the substantial price reduction should, in theory, shift relative prices noticeably and increase public transport demand. However, the extent to which this translates into lower PM2.5 concentrations remains an empirical question, as behavioural responses may be limited by inelastic demand, high car ownership, and confounding factors such as weather and economic conditions.

3.2 Empirical Literature

Empirical evidence on the environmental impacts of public transport fare subsidies is mixed, reflecting differences in policy scale, urban context, time horizons, and methodological approaches. Many studies leverage natural experiments (e.g., temporary fare reductions) and quasi-experimental methods such as difference-in-differences (DiD) or regression discontinuity designs.

Several studies report positive environmental effects. Gohl and Schrauth (2022) analyse Germany’s nationwide 9-Euro monthly ticket (a large temporary subsidy) and find a causal reduction in a composite air pollution index of over 6%, particularly in areas with strong public transport networks. Similarly, Luxembourg’s free public transport policy was associated with 5–6% reductions in road transport CO and NOx emissions (Eibinger and Sachintha 2026).

Conversely, other evaluations find limited or no air quality improvements. Albalade et al. (2024) examine Spain’s 2022 nationwide fare discounts and, using a regression discontinuity in time approach with weather and fixed effects controls, detect no statistically significant reductions across multiple pollutants (including PM and ozone). Webster (2024) evaluates Colorado’s short-term “Zero Fare for Better Air” initiative and finds no reduction in ground-level ozone; some specifications even suggest a small increase (Albalade et al. 2024; Webster 2024).

The mixed results likely stem from several mechanisms. Ridership gains are common (as observed in Queensland, with reported increases of 16-18%), but these often may reflect induced demand or substitution from non-car modes rather than large reductions in private vehicle use. Short post-treatment windows, as in the present study, may miss lagged behavioural adjustments (e.g., changes in car ownership or residential location). Measurement challenges, such as the influence of weather on pollution dispersion, seasonal patterns, and limited monitoring stations, further complicate inference.

Overall, the literature suggests that while fare subsidies can increase public transport usage and generate social benefits (affordability, congestion relief), their impact on air pollution is context-dependent and frequently smaller than proponents anticipate. Policies appear more effective when combined with complementary measures³. This underscores the value of rigorous ex-post evaluation using methods like DiD to isolate causal effects amid confounding trends.

³The U.S. Department of Transportation (2025) highlights that free or reduced-fare transit programmes and parking reforms are complementary strategies that, when implemented together, promote greater reductions in vehicle miles travelled and GHG emissions. Similarly, the National Academies of Sciences (2022) concludes that fare-free or low-fare transit policies are unlikely to generate substantial reductions in car use and emissions unless paired with measures that increase the relative cost or inconvenience of driving, such as parking pricing, congestion charging, or restrictions

4 Data & Methodology

This section describes the data and empirical approach used to assess whether Queensland’s 50-cent flat fare subsidy, introduced on 5 August 2024, reduced urban air pollution in Brisbane relative to Adelaide. The analysis focuses on fine particulate matter (PM_{2.5}) as the primary outcome, a widely used indicator in empirical economics due to its strong links to health impacts and reliable measurement⁴. The study also incorporates weather conditions and fuel prices as controls. The sample period spans 5 August 2023 to 5 August 2025 (one year pre- and post-policy).

4.1 Data

The analysis uses a station-level daily panel dataset combining air pollution, meteorological, and fuel price data for Brisbane (treated) and Adelaide (control). Descriptive statistics for all variables are provided in the Annex⁵.

Air pollution and climate data come from official government monitoring stations⁶. Brisbane data are sourced from the Department of Environment, Tourism, Science and Innovation (Queensland Government). Adelaide data are drawn from the Government of South Australia’s Open Data Portal and supplemented by the World Air Quality Index Project (AQICN). The selected stations are: Cannon Hill, Deagon, Lytton, Rocklea, Woolloongabba, Wynnum North, and Wynnum West (Brisbane); and Adelaide CBD, Elizabeth, and Le Fevre 1 (Adelaide)⁷.

Weather controls are essential in air pollution studies. Temperature (in Celsius), wind speed (in meters per second), and relative humidity (%) are calculated as daily averages, while rainfall (mm) is the daily sum of hourly observations. Fuel prices are included to isolate the policy effect from other cost-driven changes in driving behaviour. These prices have been converted from cents to Australian Dollars (AUD) per litre for easier interpretation.

Fuel price data are included as a control variable because changes in fuel costs can influence driving behaviour and, consequently, traffic-related emissions. Data are obtained from BP’s Terminal Gate Prices (TGP), reported as monthly averages in cents per litre. These prices reflect wholesale fuel costs and closely track retail fuel price movements, although they do not capture short-term retail fluctuations or retailer margins. Previous research has shown that wholesale fuel prices vary across Australian distribution centres and may exhibit different adjustment patterns over time (Burke 2025; Zhang and Burke 2020), highlighting the importance of controlling for fuel cost changes when evaluating transport-related outcomes⁸. Monthly averages are used in the daily panel, which limits within-month variation and may attenuate the estimated effect of fuel prices. Nevertheless, the use of terminal gate prices is consistent with applied research where high-frequency retail fuel data are unavailable.

on car use (National Academies of Sciences, Engineering, and Medicine 2023; U.S. Department of Transportation 2025).

⁴See, for example, Bishop and Kuminoff (2019), Chang et al. (2019), and Ebenstein et al. (2016). Previous studies have employed PM_{2.5} to examine effects on mortality, labour supply, cognitive performance, and productivity (Hanna and Oliva 2016; Chang et al. 2019; Bishop et al. 2018).

⁵The panel is unbalanced due to occasional missing observations at certain stations.

⁶Brisbane data sourced from the Department of Environment, Tourism, Science and Innovation (Queensland Government). Adelaide data sourced from the Government of South Australia Open Data Portal and supplemented by the World Air Quality Index Project.

⁷See Annex for detailed station information and data coverage.

⁸Notably, Brisbane and Adelaide have been identified as relatively competitive fuel markets compared with several other Australian cities, suggesting that fuel prices in both locations broadly reflect underlying market conditions (Valadkhani 2023).

4.2 Data Processing

Data collection involved downloading raw hourly files from the respective government portals and supplementary sources. Processing was conducted primarily in RStudio, with initial cleaning and merging performed using Excel Power Query for consistency checks. The main processing tasks included aggregating hourly observations to daily averages for PM2.5 and weather variables, and subsequently merging the pollution, weather, and fuel price datasets by date and station. The final dataset was structured as a station-day panel. To ensure data quality, missing values were handled through listwise deletion for the main specifications, and extreme outliers (physically implausible) across all variables were removed using a conservative threshold.

The distribution of raw PM2.5 concentrations was positively skewed. The dependent variable was therefore transformed using the natural logarithm, a standard practice in environmental econometrics that reduces the influence of outliers and allows coefficients to be interpreted as approximate percentage changes.

4.3 Empirical Strategy (DiD)

A difference-in-differences (DiD) approach is well-suited to this quasi-natural experiment. Brisbane implemented a sharp fare reduction on 5 August 2024, while Adelaide did not, creating a credible treated/control contrast. The method compares changes in PM2.5 levels before and after the policy in Brisbane relative to Adelaide, netting out common time trends and time-invariant differences.

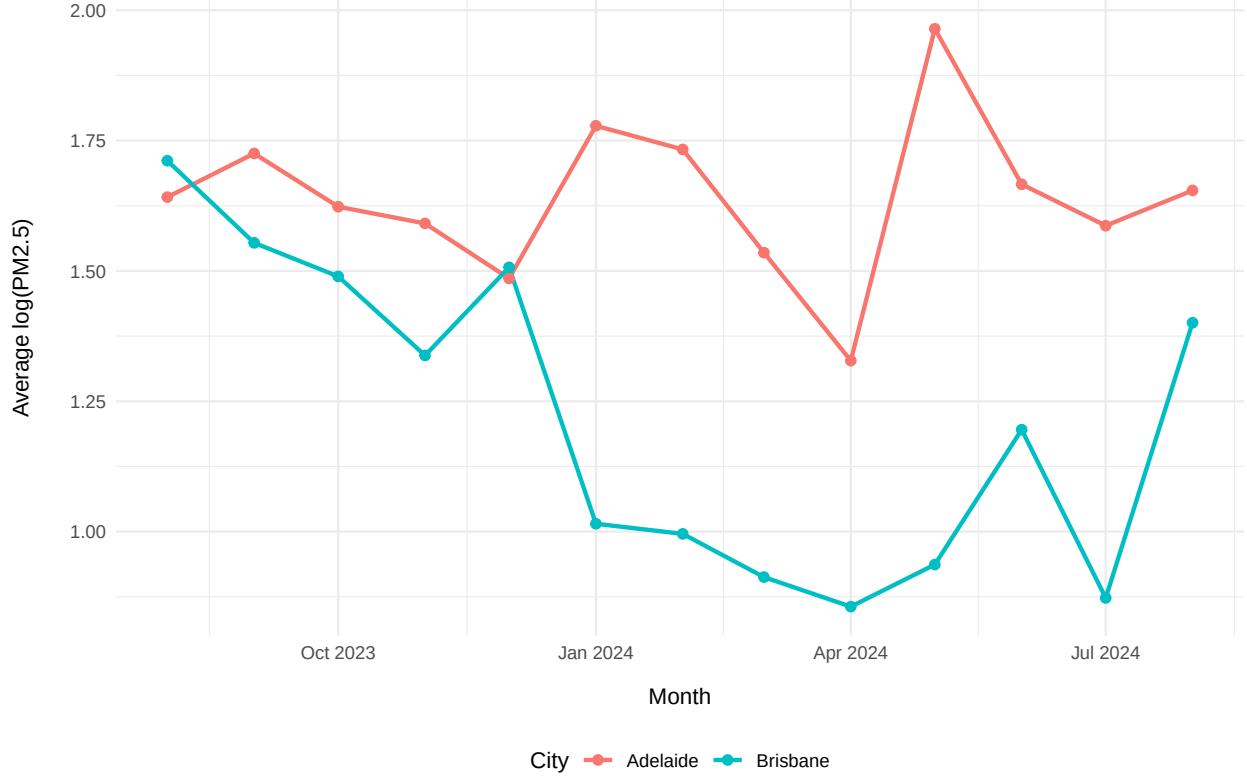
The main specification is:

$$\ln(PM2.5)_{ist} = \beta_0 + \beta_1(Treated_i \times Post_t) + \gamma X_{st} + \alpha_i + \delta_t + \epsilon_{ist}$$

where $\ln(PM2.5)_{ist}$ is the log of daily average PM2.5 concentration at station i in city s on day t ; $Treated_i \times Post_t$ is the coefficient of interest (capturing the policy effect); X_{st} is a vector of time-varying controls, including daily weather variables (temperature, wind speed, rainfall, humidity) and fuel prices; α_i are station fixed effects; δ_t are day fixed effects.

The Difference-in-Differences approach relies on several identifying assumptions. The most important is the parallel trends assumption, which states that, in the absence of the policy, PM2.5 concentrations in Brisbane and Adelaide would have evolved similarly over time. This assumption was assessed through visual inspection of Figure 1 of pre-treatment pollution trends. It can be observed that the two cities generally followed comparable movements prior to the policy, although a noticeable divergence occurred around December 2023 and January 2024. Consequently, the graphical evidence provides only partial support for the assumption and should be interpreted with caution. To mitigate concerns arising from differences in local environmental conditions, the empirical specification additionally controls for temperature, wind speed, humidity, rainfall, and fuel prices, while station and date fixed effects account for unobserved heterogeneity across monitoring locations and common shocks over time.

Figure 1: Pre-treatment PM2.5 Trends by City



A second identifying assumption is that no major Brisbane-specific shock occurred at the same time as the introduction of the fare subsidy that could independently affect air pollution. Examples include major industrial closures, environmental regulations, transport disruptions, or extreme environmental events. If such shocks occurred simultaneously with the policy, their effects could incorrectly be attributed to the subsidy. No major event of this nature was identified during the study period, reducing concerns regarding confounding policy-specific shocks.

4.4 Inference Challenges

With only 10 monitoring stations, inference requires caution. While the point estimate on the treatment interaction remains stable across specifications, statistical significance depends on how standard errors are computed. Conventional heteroskedasticity-robust errors often suggest significance, but station-level clustering (which accounts for serial correlation within stations over time) produces larger standard errors, leading to statistically insignificant coefficients in most specifications. The small number of clusters further complicates inference, as cluster-robust standard errors can be unreliable with few groups. This study therefore emphasises economic magnitude over strict statistical significance, conducts multiple robustness checks (including leave-one-out analysis), and interprets results conservatively.

5 Results

Figure 2 presents a simple before-and-after comparison of average log PM2.5 concentrations. Prior to the policy, pollution levels in Brisbane and Adelaide followed broadly similar trends. After the introduction of the 50-cent fare subsidy in August 2024, average PM2.5 levels rose in Brisbane relative to Adelaide, producing a positive raw difference-in-differences estimate. At first glance, this descriptive evidence could suggest that pollution increased following the policy intervention.

Figure 2: Before and After Comparison of PM2.5 Levels – Brisbane vs Adelaide

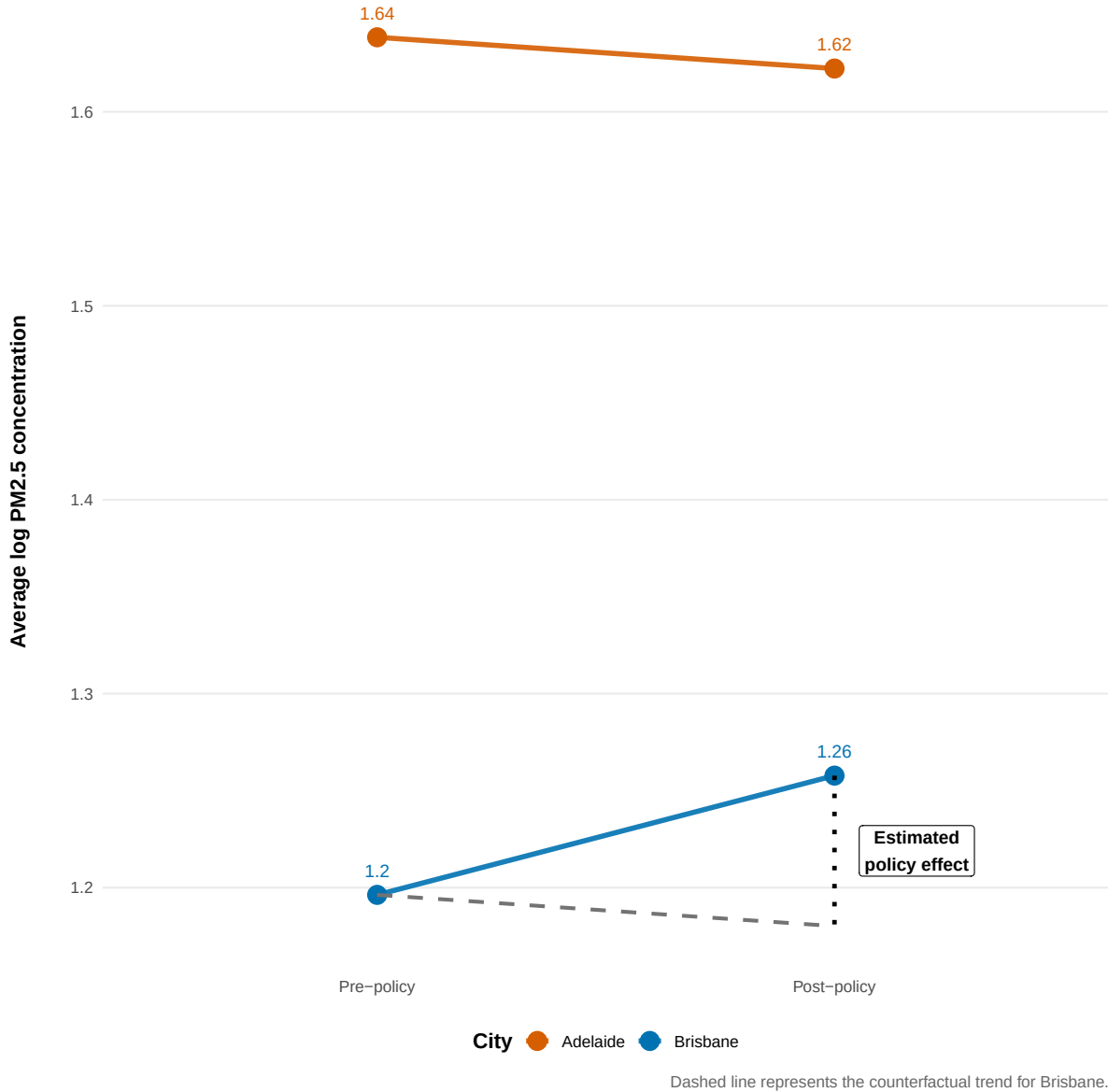


Table 1 presents the main Difference-in-Differences regression results. After controlling for weather conditions, fuel prices, station fixed effects, and date fixed effects, the estimated treatment effect

remains positive but becomes statistically insignificant. The coefficient on the interaction term between the treated city and the post-policy period is approximately 0.23, implying a positive estimated relationship between the policy and PM2.5 concentrations. However, the associated standard error is relatively large, and the estimate is not statistically distinguishable from zero at conventional significance levels. This suggests that, once relevant confounding factors are taken into account, there is no robust evidence that the fare subsidy reduced air pollution in Brisbane relative to Adelaide.

Table 1: Difference-in-Differences Regression Results

	(1)
Treated $\times$ Post (Policy Effect)	0.230 (0.174)
Temperature ($^{\circ}$ C)	0.027*** (0.004)
Wind Speed (m/s)	-0.091*** (0.022)
Humidity (%)	-0.003* (0.002)
Rainfall (mm)	-0.001 (0.001)
Fuel Price (AUD/litre)	-16.848** (5.384)
Num.Obs.	7210
R2	0.660
p < 0.1, ** p < 0.05, *** p < 0.01	

Note: The specification includes station and date fixed effects.
Standard errors are clustered at the station level.

The control variables generally display expected signs and are economically meaningful. Higher wind speeds are associated with lower pollution concentrations, consistent with atmospheric dispersion effects that reduce pollutant accumulation. Higher temperatures are associated with increased PM2.5 levels, potentially reflecting stronger atmospheric chemical reactions and pollution formation during warmer conditions. Humidity is negatively associated with pollution concentrations and is weakly statistically significant at the 10% level, suggesting that more humid conditions may contribute to lower measured particulate matter concentrations. Fuel prices are also negatively associated with pollution, consistent with the idea that higher fuel costs may reduce vehicle usage and traffic-related emissions. Rainfall displays the expected negative sign, although the estimated effect is not statistically significant in the main specification.

5.1 Robustness: Leave-One-Out Analysis

To test whether results were driven by individual stations, a leave-one-out robustness check was performed. The main model was re-estimated sequentially excluding one station at a time. The treatment coefficient remained positive and relatively stable (ranging from 0.09 to 0.32) across all specifications. No single station materially changed the overall conclusion. While estimates remained statistically insignificant under clustered standard errors, this exercise increases confidence that the main findings are not driven by outliers or idiosyncratic station-specific effects (see Table A.2 in the Annex).

5.2 Heterogeneity Analysis

An additional heterogeneity analysis was conducted by estimating station-specific post-policy effects. The results reveal substantial variation across monitoring locations, with some stations exhibiting declines in pollution levels while others showed increases or statistically insignificant changes. However, these patterns do not align systematically with treatment status. In particular, reductions in PM_{2.5} concentrations appear to occur across both treated and control locations rather than being concentrated in Brisbane following the introduction of the fare subsidy. This suggests that broader environmental or temporal factors, rather than the policy itself, may explain much of the observed variation in pollution levels during the study period.

6 Implications & Limitations

The analysis finds no statistically robust evidence that Queensland’s 50-cent flat fare policy reduced PM_{2.5} concentrations in Brisbane relative to Adelaide. These results do not imply that the policy failed in its broader goals of improving affordability and increasing public transport usage. Instead, they suggest that any environmental benefits were either too small to detect with the available data, offset by other factors, or slow to materialise.

6.1 Why Might the Policy Not Have Reduced Air Pollution?

Several explanations may account for the absence of a statistically significant reduction in PM_{2.5} concentrations following the introduction of the subsidy.

A key assumption underlying the policy is that lower public transport fares encourage commuters to switch from private vehicles to public transport. While ridership data indicate that public transport usage increased substantially after the introduction of the subsidy, increased ridership does not necessarily imply a corresponding reduction in car use. Many new public transport trips may have come from individuals who previously walked, cycled, worked from home, travelled less frequently, or made trips that would not otherwise have occurred. As a result, public transport demand can increase without generating a substantial reduction in road traffic and vehicle emissions.

Transport economics literature consistently finds that short-run travel behaviour is relatively inelastic (Litman 2023). Once households have purchased a vehicle, many of the costs of ownership become fixed, including purchase costs, insurance, registration, and maintenance. Consequently, a reduction in public transport fares may not be sufficient to induce substantial behavioural change among existing car owners. In economic terms, private vehicle users may exhibit relatively low short-run cross-price elasticity with respect to public transport fares. Individuals who have already invested in a car often continue using it because the marginal cost of each additional trip remains relatively low compared with the total cost of ownership.

Even if the policy reduced vehicle traffic to some extent, the resulting effect on air pollution may have been small relative to other determinants of PM_{2.5} concentrations. The regression results indicate that weather conditions play an important role in explaining pollution levels. Wind speed, temperature, humidity, and rainfall all influence the formation, dispersion, and removal of airborne particles. In addition, pollution concentrations can be affected by industrial emissions, port activities, construction activity, regional pollution transport, seasonal factors, and natural events such as bushfire smoke. These influences may be substantially larger than any short-run reduction in vehicle emissions generated by the fare subsidy.

Behavioural responses to transport policies often occur gradually rather than immediately. Households may require time to adjust commuting habits, relocate closer to public transport, change workplace travel arrangements, or alter vehicle ownership decisions. The post-treatment period in this study covers approximately one year following implementation. While sufficient to capture immediate responses, it may be insufficient to observe longer-term behavioural changes that could eventually affect air quality.

The fare subsidy was initially introduced as a six-month trial before later being made permanent. During the trial period, commuters may have been uncertain about whether the reduced fares would remain in place. Economic theory suggests that individuals are less likely to make substantial behavioural adjustments when a policy is perceived as temporary. For example, a commuter may be reluctant to change long-established travel habits if the subsidy could be removed after only a few months. Consequently, uncertainty regarding the future of the policy may have weakened its initial impact.

An additional possibility is that increased demand for public transport generated crowding effects. If trains and buses became more congested, some potential users may have chosen not to switch from private vehicles despite the lower fares. Although Queensland reported substantial increases in ridership following the policy, the extent to which existing infrastructure and service frequency were able to accommodate this demand remains unclear. If capacity constraints emerged, the environmental benefits of the subsidy may have been smaller than anticipated.

6.2 Limitations

Several limitations should be considered when interpreting the findings of this study.

The most important limitation concerns statistical inference. Although the dataset contains more than 7,000 daily observations, the effective number of clusters is only ten monitoring stations.

The policy was introduced on 5 August 2024, and the analysis extends until 5 August 2025. This provides only one year of post-treatment data. While sufficient to capture immediate impacts, a longer observation period may be required to identify medium- or long-term environmental effects. Changes in commuting behaviour, vehicle ownership, and transport demand often occur gradually and may not be fully reflected within a single year.

Although the model controls for weather conditions, fuel prices, station fixed effects, and date fixed effects, other factors affecting air pollution may remain unobserved. Examples include local industrial activity, construction projects, changes in freight transport, shipping activity, regional pollution events, and other policy interventions. If these factors differ between Brisbane and Adelaide during the study period, the estimated treatment effect may be biased.

PM_{2.5} is widely used as an indicator of air quality and has important health implications. However, it captures only one dimension of pollution. Transport emissions also affect nitrogen dioxide (NO₂),

carbon monoxide (CO), ultrafine particles, and greenhouse gas emissions. Consequently, the absence of a detectable PM_{2.5} effect does not necessarily imply the absence of environmental benefits more broadly.

Data availability imposed several restrictions on the analysis. A number of monitoring stations originally considered for inclusion were excluded due to incomplete pollution or weather records. This reduced the number of available stations and may limit the representativeness of the final sample. In addition, weather data for Adelaide had to be compiled from multiple sources, increasing the complexity of data preparation and the potential for measurement inconsistencies.

The final dataset contains seven monitoring stations in Brisbane but only three in Adelaide. Although station fixed effects help account for persistent differences between locations, the imbalance means that Brisbane contributes a larger share of observations to the analysis. This may influence the precision of estimates and reduce comparability between treatment and control groups.

Fuel prices were included as an important control variable because fuel costs influence driving behaviour and transport demand. However, the available data consisted of monthly average BP Terminal Gate Prices rather than daily retail fuel prices faced by consumers. These prices measure wholesale fuel costs and therefore do not fully capture retail margins, local pricing differences, short-term price fluctuations, or differences across fuel retailers. Furthermore, the use of monthly fuel prices in a daily regression framework limits within-month variation. As a result, the estimated fuel price effect should be interpreted as capturing broader fuel cost trends rather than precise day-to-day changes in consumer fuel prices.

Taken together, these limitations suggest that the findings should be interpreted as evidence that no statistically robust reduction in PM_{2.5} concentrations can be detected within the available data, rather than definitive evidence that the fare subsidy generated no environmental benefits whatsoever. Additional years of post-policy data, a larger number of monitoring stations, and alternative pollution indicators would help provide a more comprehensive assessment of the policy's environmental impact.

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Appendix

A.1 Descriptive Statistics

Table 2: Descriptive Statistics

	N	Mean	SD	Min	Max
log(PM2.5)	7223	1.35	0.73	-2.30	3.37
Temperature	7219	20.16	4.68	6.60	35.68
Wind Speed	7212	2.42	1.39	0.24	12.61
Humidity	7223	67.59	12.62	17.28	100.00
Rainfall	7221	2.17	8.42	0.00	203.60
Fuel Price	7223	1.69	0.10	1.53	1.91

Note: While physically impossible, negative $PM_{2.5}$ values commonly occur due to instrument noise and measurement error around the zero baseline. Environmental protection agencies (including the US EPA and Australian networks) expect these readings. Regulatory guidelines advise against deleting valid negative data, as removing them skews the dataset and artificially inflates the perceived level of air pollution(Jiang et al. 2023).

A.2 Air Quality Monitoring Stations

The analysis uses data from 10 air quality monitoring stations located across the Brisbane and Adelaide metropolitan areas. The selected stations represent a mix of urban, suburban, industrial, and port-related environments, allowing the dataset to capture different sources of particulate pollution. Several other stations were excluded from the analysis because of insufficient data availability, and only stations with sufficiently complete observations throughout the study period were retained.

Brisbane

- **Cannon Hill:** Located in eastern Brisbane, Cannon Hill is a predominantly urban and residential area influenced by local traffic and commercial activity.
- **Deagon** is situated in northern Brisbane, Deagon is a suburban location affected by both residential traffic and broader regional air flows.
- **Lytton** is an industrial and port precinct near the Port of Brisbane, with pollution levels potentially influenced by freight transport and industrial operations.
- **Rocklea** is a mixed residential and industrial suburb in southern Brisbane, located near several major transport corridors.
- **Woolloongabba** is an inner-city station close to Brisbane’s central business district, Woolloongabba is strongly influenced by high traffic volumes and dense urban activity.
- **Wynnum North** is a suburban station in Brisbane’s eastern coastal area, reflecting pollution conditions in a mainly residential environment.
- **Wynnum West** is also located in eastern Brisbane, Wynnum West represents a suburban setting with lower traffic intensity than the inner-city locations.

Adelaide

- **Adelaide CBD** is a station located in the city centre and captures pollution associated with dense urban development and commuter traffic.
- **Elizabeth** is a major northern suburb of Adelaide characterised by mixed residential and industrial land use.
- **Le Fevre 1** is located on the Le Fevre Peninsula, this station monitors air quality in an area associated with port activities, heavy industry, and freight transport.

Choice of Unit of Analysis: City Average vs Station-Level Panel

A key decision was whether to aggregate all monitoring stations into a single city-level daily value or to maintain station-level observations. This study adopts the latter approach, preserving spatial variation in pollution levels within each metropolitan area.

Keeping individual stations substantially increases the number of observations and statistical power. Air pollution, particularly PM_{2.5}, varies significantly across locations due to differences in traffic density, industrial activity, and proximity to major roads. Aggregating stations would mask this variation and introduce measurement error.

The station-level structure also enables the inclusion of station fixed effects, which control for time-invariant local characteristics (e.g., proximity to highways or ports). This approach is standard in environmental economics literature, as it exploits both spatial and temporal variation in air quality data.

The final dataset is therefore structured as a station–day panel, with each observation representing daily pollution and weather conditions at a specific monitoring station.

A.3 Robustness Checks (Leave-One-Out)

Table 3: Leave-One-Out Robustness Check

	Station Removed	Coefficient	Std. Error
Adelaide CBD.treated:post	Adelaide CBD	0.286	0.220
Cannon Hill.treated:post	Cannon Hill	0.266	0.183
Deagon.treated:post	Deagon	0.267	0.183
Elizabeth.treated:post	Elizabeth	0.324	0.188
Le Fevre 1.treated:post	Le Fevre 1	0.087	0.120
Lytton.treated:post	Lytton	0.150	0.161
Rocklea.treated:post	Rocklea	0.255	0.185
Woolloongabba.treated:post	Woolloongabba	0.259	0.186
Wynnum North.treated:post	Wynnum North	0.239	0.190
Wynnum West.treated:post	Wynnum West	0.173	0.175

Figure 3: Leave-One-Out Robustness Check - Policy Effect by Station Removed

